

Research Matrix Analysis

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Component	Thesis 1	Thesis 2	Thesis 3	Thesis 4	Thesis 5	Thesis 6
Undergraduate/Master's/Doctorate	Undergraduate	Undergraduate	Undergraduate	Undergraduate	Undergraduate	Doctorate
Title/Author/Year	KUMPAS: AI RECOGNITION OF COMMON SIGN LANGUAGE FOR DEAF PEOPLE TO DECREASE LANGUAGE BARRIERS USING CONVOLUTIONAL NEURAL NETWORK ALGORITHM Anora, Jerick F. Brusas, Rose Anne Mae A. Garcia, MA. Michaela C. Miguel, Sean	A Comparative Study of Object Detection Algorithms in Providing Assistance for Visually Impaired Person De Leon, Yuan Raphael T. Ilarde, Dean Patrick Maranan, Christopher Jr G. Papellero, Jobet Adrian Pastoral, Jommel R.	DEVELOPMENT OF HANDYDRAWING: A BASIC VIRTUAL DRAWING APPLICATION USING MEDIAPIPE ALGORITHM Bacong, Jefferson A. Combras, John Joshua B. Magpayo, Jade Irvine D. Bacong, Jefferson A. Combras, John Joshua	KONEKTA: MOBILE APPLICATION FOR ASSISTANCE OF VISUALLY IMPAIRED USING OBJECT DETECTION AND TEXT-TO-SPEECH De Leon, Yuan Raphael T. Ilarde, Dean Patrick Maranan, Christopher Jr. G. Papellero, Jobet Adrian	Development of KUMPAS: A Mobile Application for Filipino sign language recognition utilizing convolutional neural network algorithm Anora, Jerick F. Brusas, Rose Anne Mae A. Garcia, MA. Michaela C. Miguel, Sean 2023	Intelligent agent using artificial neural network for e-service of bureau of fire protection Mata, Khatalyn Espiritu 2018

	Justine M. 2023	2023	B. Magpayo, Jade Irvine D. 2023	Pastoral, Jommel R. 2023		
Institution	University of Makati	University of Makati	University of Makati	University of Makati	University of Makati	University of the East
Research Problem/Objectives	Produce real-time translation of Filipino sign language to ease the problem of deaf Filipino people	Compare the performance of two popular object detection algorithms, YOLOv4 Tiny and SSD, for their suitability in applications for visually impaired people	How to design, develop, and implement a web-based basic virtual drawing application called HandyDrawing that utilized MediaPipe algorithms to detect open and close hand gestures, enabling children with amputated fingers to perform drawing functions which may provide a user-friendly interface that promotes inclusivity in performing drawing	Create a feature for a mobile application that can recognize objects and provide voice outputs based on what it finds	Develop a Filipino sign language detector for deaf people utilizing CNN.	Design a system with intelligent agents that can communicate and provide accurate information to the applicant of fire safety certificate for the bureau of fire protection in the National Capital Region.

			activities specific to their hand condition			
Theoretical/Conceptual Framework	Input: - Dataset Collection - Images - Android Studio - Java - PyCharm - Python Process: - Data Gathering - Implementation of CNN Algorithm - Evaluation Procedure Output: - Kumpas application	Input: - Images of indoor objects - Videos of indoor objects Process: - Pre-trained Single Shot Detector (SSD) Algorithm - Pre-trained You Only Look Once (YOLO) algorithm - Comparing of FPS	N/A	Input: - Visual Impairment - Assistive Device - Mobile Applications - Algorithm used for development - Programming tools used for Development - ISO 25010 Process: - Design Development Testing Output: - Konekta: A mobile	N/A	Input: - Fire Safety Certificate Applicant information - Bureau of Fire Personnel information - Mobile Device - Computer Kiosk Process: - Voice Recognition - Image Recognition - Text Output: - Notification

		and accur acy Output: - A Comp arativ e Study of Object Detect ion Algorit hms in Provid ing Assist ance for Visuall y Impair ed Perso n		applica tion for assista nce of the visuall y impair ed in Object Detecti on and text-to- speec h		- Sched uling
Research Design	Experimental research design	Experimental and applied studies	Cross Industry Standard Process for Data-Mining (CRISP-DM)	Developmenta l and descriptive approach, which include surveys, observational studies, and case studies. The data collected may be	Experimental research design	Descriptive Research Design

				quantitative or qualitative.		
Participants/Sample	400 images per phrase	Sample images and videos of objects are 20-30 objects	A female child who has congenital hand condition that affects her index, middle, ring, and pinky fingers captured from various angles and positions. As a result, a total of 24,000 sets of 21 hand landmarks were obtained from the dataset	15 visually impaired individuals 15 IT experts 30 total respondents	Respondents were chosen at random, respondents were divided into two groups: signers and non-signers.	Unknown number of members of the Bureau of Fire Protection
Data Collection Methods	Collected customized dataset with at least 400 images per phrase	Collection of reliable internet sources, COCO datasets, and online literature	Self collected images for hand landmarks	Surveys and observational studies	Surveys and Likert's scale evaluation	Interview, questionnaire and evaluation form in data gathering.
Instrument Used	Python - PyCharm Java - Android Studio	Python - PyCharm Jupyter Notebook	Python - PyCharm MediaPipe OpenCV TensorFlow NumPy	Android Studio Chipmunk Google Colab Java SDK 17 Anaconda 3 Python 3.9	Python - PyCharm Java - Android Studio Android SDK	Frontend: Bootstrap, HTML, CSS, JavaScript, JQuery

					21: Android 5.0(Lollipop) Lite CV2 NumPy Adobe XD Mobile Camera (At least 8 mp)	Backend: CodeIgniter(PHP) MySQL Tracking.JS iTextMo API
Data Analysis Techniques	Convolutional Neural Network	Evaluated algorithm's accuracy and speed through Average Precision, Intersection over Union, and Frames per Second	Used to track 21 hand landmarks across 24,000 datasets Standards of the ISO25010	Standards of the ISO25010 and Linkert's Principle	The application's performance is tested based on the criteria of ISO 25010 software quality model.	Likert Scale
Key Findings	Convolutional Neural Network Algorithm is an effective method for Filipino Sign language recognition, with the highest accuracy being 83% for "Sandali" and lowest at 72% (Ako)	YOLOv4 outperforms SSD in terms of accuracy (Intersection over Union and Average Precision) and speed (Frames per Second) as it only utilizes a single CNN to predict bounding boxes and	1. Enables children with congenital hand conditions to draw virtually by detecting open/close	The researchers developed Konekta: A mobile application for assistance of the visually impaired in Object Detection and Text to speech and were able to implement the main features using	1. Kumpas was successfully developed using android studio as the integrated development environment. 2. The researchers program meets ISO 25010 quality standards.	- The intelligent agent using ANN for the E-service of BFP for the processing of the application for fire safety certificates is absolutely necessary. It enhanced the effectiveness and efficiency

		class scores, while SSD uses multiple CNNs	hand gestures 2. Successfully implemented a gesture-based drawing tool using Media Pipe.	Android Studio using Java and Python. The application was limited to the objects under the COCO dataset. 62 out of the 80 objects were successfully detected. 46 out of 62 objects got at least 70% accuracy	3. Researchers created a survey for 15 signers and 15 non-signers to assess the acceptance of the application that matches the feature criteria. The game was able to achieve positive remarks by the respondents using the software quality model ISO 25010 which was based on functional suitability, reliability, portability, performance efficiency and suitability.	of serving the fire safety certificates applicants. - Comparatively, the system with intelligent agents using artificial neural network is better in terms of its effectiveness compared to the existing process for the application of fire safety certificates. - the system with intelligent agents using artificial neural networks for the e-service for BFP was found very effective. It minimized the number of processes in applying fire safety certificates.
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Limitations/Gaps	Kumpas only contains 13 Filipino sign languages	Only 20-30 types of objects are gathered, and are analyzed using pre-trained models to save on time and hardware cost	• Limited to Desktop/Laptop devices • Recommended configurations require hardware that may not be easily accessible to certain parties • only recognizes learned hand gestures • shows colors based on	Objects from the COCO dataset were the only objects that can be detected, lighting may affect the application's ability to detect as well as partially buried objects, objects moving quickly may also not be detected Android 10 devices and under may access Unavailable for IOS	Only supports static sign languages. Sign languages that are dynamic are not included in the study Kumpas is only available without internet. Kumpas is only downloadable on Android mobile devices Minimum system requirements are: 4GB RAM, 128 ROM, Android 10 or up with at least 8-megapixel camera. Can only recognize signs with a	- This study focused on designing systems with intelligent agents as a new front liner of the bureau of fire protection and personal assistant of applicants in applying fire safety certificates. the system is able to process data available upon its installment and other forthcoming data. data obtained from the previous years are not included in the system. It includes processing applications for fire safety certificates, records keeping, scheduling of fire inspector
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			their exact equivalents in the color picker which might affect colorblind users • Device hardware can affect performance • Cannot recognize a hand gesture if all fingers have been amputated		good lighting	and notifications. - on the notification aspect of the system, the sms feature is limited to the sending of messages to applicants. It cannot validate whether the messages were received and have no capability to accept reply messages. the system does not include the identification of irresponsible applicants which refers to those duly notified as to unpaid fees renewable certificates and delinquents in the retrieval of the certificates which are
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						ready for release
Relevance to your Study	Hand gesture tracking and AI training models	Determining which algorithm is best for aiding visually impaired people	Hand gesture tracking and AI training models	Object detection capabilities of Android devices	Hand gesture tracking and AI training models	Voice controls with the use of Artificial Intelligence